

Trim, Thrust, and Power Analysis of a Cropped Delta Wing UAV

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1 Problem Description

1.1 Introduction and Abstract

Aircraft performance analysis is essential to understand how aerodynamic forces and propulsion requirements vary with different flight conditions. For unconventional configurations such as cropped delta reflex wings, analytical solutions are complex, and numerical methods become necessary. This project presents the trim analysis and performance evaluation of a cropped delta wing unmanned aerial vehicle (UAV) under steady level flight conditions. By applying numerical techniques, we evaluate the equilibrium state where the aircraft experiences zero pitching moment, allowing for a detailed study of propulsion and aerodynamic efficiency.

1.2 Objectives

The primary objective of this project is to analyze the trim condition, thrust required, and power required of the UAV using given aerodynamic and geometric parameters. Specifically, the project aims to:

- Perform a comprehensive trim analysis to ensure zero pitching moment ($C_m = 0$).
- Develop an iterative MATLAB-based numerical solver to evaluate flight mechanics.
- Establish the mathematical relationship between angle of attack (α) and flight velocity (V).
- Compute the thrust required (T_r) and power required (P_r) for steady level flight.
- Evaluate key performance metrics, including lift-to-drag ratio (C_L/C_D) and the endurance parameter ($C_L^{3/2}/C_D$).

1.3 Methodology and Scope

The methodology involves developing a MATLAB model that varies the angle of attack from 0° to 12° . For each increment, the model computes trim elevator deflection, lift/drag coefficients, and power requirements. This study bridges aerodynamics, flight mechanics, and numerical programming to identify key performance trends, such as the minimum power condition for maximum endurance.

2 Geometry and Aerodynamic Modeling

The UAV configuration is a wing-alone cropped delta wing. The calculated and given parameters are as follows:

- **Configuration:** Cropped delta wing (wing-alone)

- **Root Chord (C_r):** 0.9 m
- **Tip Chord (C_t):** 0.15 m
- **Span (b):** 1.5 m
- **Oswald Efficiency (e):** 0.89
- C_{D0} : 0.03
- C_{m0} : 0.01
- $C_{L\alpha}$: 2.92 per rad
- $C_{L\delta_e}$: 0.265
- $C_{m\alpha}$: -0.292
- $C_{m\delta_e}$: -0.4
- **UAV Mass:** 3.5 kg (Weight $\mathbf{W} = 35$ N, as **Gravity** $g = 10$ m/s²)
- **Air density (sea level) (ρ):** 1.225 kg/m³

3 Theory and Formulation

3.1 Trim Condition

For steady trimmed flight, the pitching moment coefficient must be zero:

$$C_m = C_{m0} + C_{m\alpha}\alpha + C_{m\delta_e}\delta_e = 0 \quad (1)$$

From this, the trim elevator deflection ($\delta_{e,trim}$) is computed. Also, the lift coefficient is given by:

$$C_L = C_{L0} + C_{L\alpha}\alpha + C_{L\delta_e}\delta_e \quad (2)$$

3.2 Level Flight Relations

Level flight equilibrium requires lift to equal weight:

$$L = W = \frac{1}{2}\rho V^2 S C_L \quad (3)$$

Thrust required (T_r) and Power required (P_r) are derived from the drag polar(C_D):

$$C_D = C_{D0} + k C_L^2 \quad (4)$$

$$T_r = D = \frac{1}{2}\rho V^2 S (C_{D0} + k C_L^2) \quad (5)$$

$$P_r = T_r V \quad (6)$$

3.3 Pre-processing Calculations

The following geometric parameters were computed to initiate the MATLAB iterative procedure:

1. Taper Ratio (λ):

$$\lambda = \frac{c_t}{c_r} = 0.1667 \quad (7)$$

2. Planform Area (S):

$$S = \frac{b}{2} C_r (1 + \lambda) = 0.7875 \text{ m}^2 \quad (8)$$

3. Aspect Ratio (AR):

$$AR = \frac{b^2}{S} = 2.8571 \quad (9)$$

4. Induced Drag Correction Factor (k):

$$k = \frac{1}{\pi e AR} = 0.1251 \quad (10)$$

3.4 MATLAB Algorithm

```

1 %% Aircraft Design Project 4: Cropped Delta Wing UAV Performance
  Analysis
2 % This script computes the trim condition and performance metrics for a
3 % cropped delta wing UAV and generates plots against velocity and alpha.
4
5 clear; clc; close all;
6
7 %% 1. Given Parameters and Constants
8 m = 3.5;           % UAV mass [kg]
9 g = 10;            % Gravity [m/s^2]
10 W = m * g;         % Weight [N]
11 rho = 1.225;        % Air density at sea level [kg/m^3]
12
13 % Geometry
14 Cr = 0.9;           % Root chord [m]
15 Ct = 0.15;          % Tip chord [m]
16 b = 1.5;            % Span [m]
17 e = 0.89;           % Oswald efficiency factor
18
19 % Aerodynamic Coefficients
20 CD0 = 0.03;         % Zero-lift drag coefficient
21 Cm0 = 0.01;         % Pitching moment at zero lift
22 CL_alpha = 2.92;    % Lift curve slope [per rad]
23 CL_de = 0.265;      % Lift due to elevator deflection
24 Cm_alpha = -0.292;  % Pitching moment due to alpha
25 Cm_de = -0.4;       % Pitching moment due to elevator deflection
26
27 %% 2. Preprocessing Calculations
28 lambda = Ct / Cr;    % Taper ratio
29 S = (b/2) * Cr * (1 + lambda); % Planform area [m^2]
30 AR = b^2 / S;        % Aspect ratio

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```

31 k = 1 / (pi * e * AR); % Induced drag correction factor
32
33 %% 3. Iterative Trim Procedure
34 alpha_deg = 0:0.5:12; % Angle of attack range [deg]
35 alpha_rad = deg2rad(alpha_deg); % Convert to radians
36
37 % Pre-allocate arrays for efficiency
38 de_trim = zeros(size(alpha_rad));
39 CL_trim = zeros(size(alpha_rad));
40 V_flight = zeros(size(alpha_rad));
41 CD_trim = zeros(size(alpha_rad));
42 Tr = zeros(size(alpha_rad));
43 Pr = zeros(size(alpha_rad));
44
45 for i = 1:length(alpha_rad)
46     a = alpha_rad(i);
47
48     % a. Trim elevator deflection (Cm = 0 condition)
49     % Cm0 + Cm_alpha*alpha + Cm_de*de = 0
50     de_trim(i) = -(Cm0 + Cm_alpha * a) / Cm_de;
51
52     % b. Trim lift coefficient
53     CL_trim(i) = CL_alpha * a + CL_de * de_trim(i);
54
55     % c. Flight velocity for steady level flight (L=W)
56     V_flight(i) = sqrt((2 * W) / (rho * S * CL_trim(i)));
57
58     % d. Drag coefficient
59     CD_trim(i) = CD0 + k * (CL_trim(i)^2);
60
61     % e. Thrust required (Tr = D)
62     Tr(i) = 0.5 * rho * (V_flight(i)^2) * S * CD_trim(i);
63
64     % f. Power required (Pr = Tr * V)
65     Pr(i) = Tr(i) * V_flight(i);
66 end
67
68 % Phase 2: Performance Metrics
69 LD = CL_trim ./ CD_trim; % Lift-to-Drag ratio
70 CL32D = (CL_trim.^1.5) ./ CD_trim; % Endurance parameter
71
72 %% 4. Visualization
73
74 % --- Page 1: Plots against Flight Velocity ---
75 figure('Name', 'Variation with Flight Velocity', 'Units', 'normalized',
76     'Position', [0.1, 0.1, 0.8, 0.8]);
77 plot_data = {alpha_deg, rad2deg(de_trim), CL_trim, CD_trim, Tr, Pr, LD,
78     CL32D};
79 titles = {'Trim Angle of Attack', 'Trim Elevator Deflection', 'Lift
80     Coefficient', ...
81     'Drag Coefficient', 'Thrust Required', 'Power Required', ...
82     'C_L/C_D', 'C_L^{3/2}/C_D'};
83 y_labels = {'\alpha (deg)', '\delta_{e,trim} (deg)', 'C_{L,trim}', 'C_D',
84     'T_r (N)', 'P_r (W)', 'E_f', 'E_p'};
85
86 for j = 1:8
87     subplot(4, 2, j);

```

```

84     plot(V_flight, plot_data{j}, '-o', 'MarkerSize', 3, 'LineWidth', 1);
85     title(titles{j});
86     xlabel('Flight Velocity (m/s)');
87     ylabel(y_labels{j});
88     grid on;
89 end
90
91 % --- Page 2: Plots against Angle of Attack ---
92 figure('Name', 'Variation with Angle of Attack', 'Units', 'normalized',
93       'Position', [0.1, 0.1, 0.8, 0.8]);
94 for k = 1:8
95     subplot(4, 2, k);
96     % Note: x-axis here is alpha_deg
97     if k == 1
98         % Plotting V vs alpha for the first plot on this page
99         plot(alpha_deg, V_flight, '-o', 'MarkerSize', 3, 'LineWidth', 1)
100     ;
101         title('Flight Velocity');
102         ylabel('V (m/s)');
103     else
104         plot(alpha_deg, plot_data{k}, '-o', 'MarkerSize', 3, 'LineWidth',
105       , 1);
106         title(titles{k});
107         ylabel(y_labels{k});
108     end
109     xlabel('Angle of Attack (deg)');
110     grid on;
111 end

```

Listing 1: MATLAB Code for UAV Trim and Performance Analysis

4 Results and Discussion

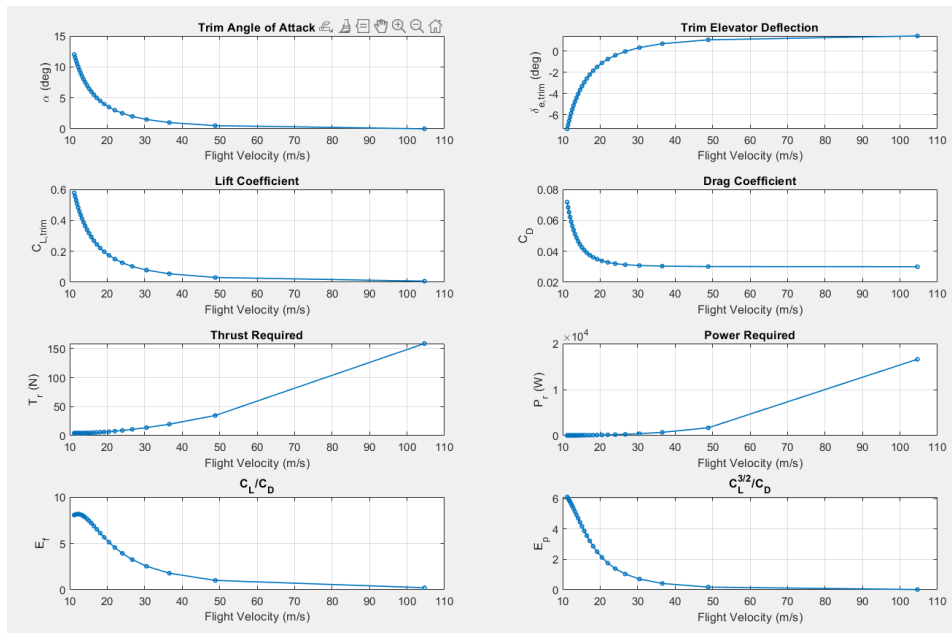


Figure 1: Plots against Flight Velocity

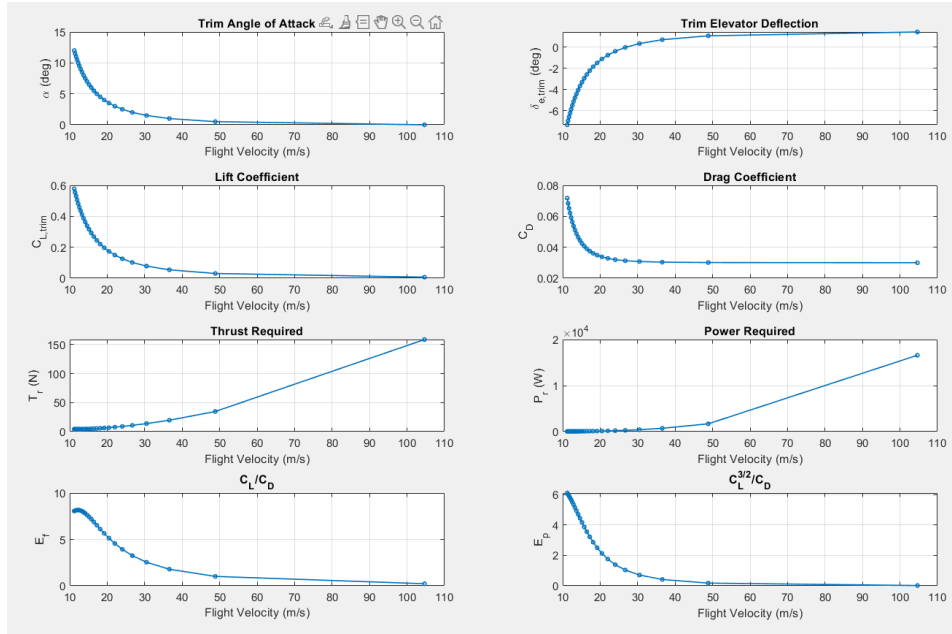


Figure 2: Plots against Angle of Attack

The results of the numerical analysis describe the aerodynamic and propulsive behavior of the cropped delta wing UAV across the tested flight envelope (0° to 12° angle of attack).

4.1 Trim Characteristics

The trim analysis ensures the UAV maintains a steady level flight without rotating about its lateral axis ($C_m = 0$).

- **Elevator Deflection ($\delta_{e,trim}$):** As velocity increases, the required angle of attack (α) for lift equilibrium decreases. Consequently, the elevator deflection must adjust linearly with α to counter the inherent pitching moment of the wing.
- **Velocity vs. Alpha:** A clear inverse relationship is observed; at high angles of attack, the UAV flies at lower speeds, whereas high-speed flight requires a lower, more efficient angle of attack.

4.2 Power and Thrust Requirements

The thrust and power curves provide insight into the UAV's energy consumption.

- **Thrust Required (T_r):** The thrust curve exhibits a "U-shape." At low speeds, induced drag dominates (kC_L^2), while at high speeds, parasitic drag (C_{D0}) becomes the primary factor.
- **Power Required (P_r):** The minimum point of the power curve represents the condition for Maximum Endurance. This occurs at a lower velocity than the minimum thrust condition.

4.3 Aerodynamic Efficiency

Performance metrics were evaluated to identify optimal flight regimes.

- **Lift-to-Drag Ratio (C_L/C_D):** The peak of this curve identifies the angle of attack for Maximum Range. For this cropped delta configuration, the maximum efficiency is achieved when zero-lift drag and induced drag are balanced.
- **Endurance Parameter ($C_L^{3/2}/C_D$):** The maximum value of this parameter corresponds to the speed where the UAV can remain airborne for the longest duration on a single battery charge.

5 Presentation Slides

The drive link for the presentation slides is provided here:

1 <https://drive.google.com/file/d/1xxucilKH3wmfBtIV085ckmnonnA-R1lV/view?usp=sharing>

6 Conclusion

This project successfully demonstrated the iterative numerical methodology required to analyze a cropped delta wing UAV. The study yielded several critical insights:

- **Coupled Constraints:** Flight performance is not determined by aerodynamics alone but by the coupling of lift, drag, and pitch trim requirements.
- **Trim Impact:** The elevator deflection required for trim ($\delta_{e,trim}$) significantly alters the total lift coefficient, emphasizing the need for trim analysis in early design phases.
- **Numerical Necessity:** Because the relationship between α and V is non-linear in steady level flight, numerical tools like MATLAB are essential for identifying optimal flight envelopes.